

The Confidence Chain: A Formal Framework for Cross-Framework Scientific Evaluation and a Proposal for Explicit Epistemic Provenance in Peer Review

Patrick McCarthy

2025

Abstract

Scientific citation practice carries an implicit assumption that when Paper B cites Paper A, Paper B inherits Paper A’s credibility for the specific claim cited. This assumption is never made explicit, never quantified, and critically — never revalidated as the citation propagates through the literature. We call this the confidence chain problem. We propose a formal solution: explicit epistemic provenance using a knowledge graph schema in which every proposition carries a provenance triple (attribution, evidence, derivation) and a computable confidence score $C_1(p)$. Building on this schema, we introduce the Decomposition Theorem: given a set of frameworks sharing base assumptions (equivalency classes), every derivation is classifiable as a collapse (independently derivable from shared premises), a contradiction (derivation conflicts with another framework from the same premises), or a novel contribution. We demonstrate this method by applying it to a formal theory of intelligence (uncertainty bounding) against fifteen prior frameworks including Schmidhuber’s formal theory of creativity, Friston’s free energy principle, Shannon’s information theory, and Peirce’s pragmatic epistemology. The analysis produces ten equivalency classes, four confirmed overlap propositions, five novel contributions, and three emergent propositions not present in any individual framework — confirming the method’s ability to discover new knowledge at the intersection of independent projections. We close by proposing a concrete peer review standard based on explicit confidence chains, equivalency class assignment, and inter-framework contradiction checking, and argue that the current review process is necessary but structurally insufficient for the epistemic claims it is asked to validate.

The Confidence Chain: A Formal Framework for Cross-Framework Scientific Evaluation and a Proposal for Explicit Epistemic Provenance in Peer Review

1. Introduction

Every citation is an implicit confidence transfer.

When a paper states “as shown in [A], proposition P holds,” the reader infers: the authors believe P is true, they believe [A] established P, and the peer review process that accepted the current paper did not find this belief unreasonable. The confidence the reader attaches to P — absent any independent verification — derives from this chain of inference. That chain is never made

explicit. The probability that P is actually true, conditional on the full evidence base behind $[A]$, is not stated. The specific claim from $[A]$ being invoked (papers contain many claims) is often not identified. Whether the authors have independently verified P , or merely assumed it from the citation, is not recorded. And once the paper is published and itself cited, the confidence in P propagates further, diluting its connection to the original evidence with each step.

We call this the *confidence chain problem*. It has three components:

The attribution-confidence conflation. Citation assigns attribution (this claim came from $[A]$) but is read as confidence (this claim is as reliable as $[A]$ itself). These are different. Attribution is a factual record; confidence is an epistemic judgment. Current practice encodes only attribution.

The revalidation gap. Once $[A]$ establishes P and $[B]$ cites $[A]$ for P , P 's confidence level in $[B]$'s readers' minds is determined by $[A]$'s general reputation, not by the specific evidence behind P in $[A]$. If $[A]$'s evidence for P later weakens — a failed replication, a revised assumption, a boundary condition found — $[B]$ and every paper in $[B]$'s citation chain retains its inherited confidence in P with no automatic update.

The cross-framework blindness. The current review process validates a paper against itself: is it internally consistent, is it novel, does it engage with the relevant literature? It does not systematically check whether the paper's claims collapse into existing results from shared premises (in which case they are confirmations, not contributions), contradict existing results from shared premises (in which case one of the frameworks is wrong or misscoped), or derive genuinely new results that no prior framework reaches from the same starting points. These three outcomes — collapse, contradiction, novelty — are not currently distinguished by the review process.

This paper proposes one possible approach to addressing all three components, implemented as a working knowledge graph. We want to be clear about the spirit of this proposal: it is not a critique of peer review or of the researchers who work within it. The current system has produced extraordinary science for centuries. What we are proposing is an extension — a way to make explicit what the current system leaves implicit — offered as a starting point for community discussion, not as a finished standard. The specific measures, definitions, and procedures we describe below are our best attempt at a concrete starting point. We expect — and welcome — the community to propose better ones.

The paper is organized as follows. Section 2 develops the formal framework: provenance triples, confidence scores, equivalency classes, and the Decomposition Theorem. Section 3 presents the knowledge graph schema. Section 4 demonstrates the framework on a running example — a formal theory of intelligence evaluated against fifteen prior frameworks. Section 5 presents the results: equivalency classes, confirmed overlaps, novel contributions, and emergent propositions. Section 6 establishes the confidence profile of the demonstrated theory. Section 7 proposes the new peer review standard. Section 8 discusses scaling to the full literature and the emergent-knowledge prediction. Section 9 concludes.

2. The Formal Framework

2.1 Provenance Triples

Definition 1 (Proposition). A proposition p is a claim that admits a truth value relative to a model. Propositions are the atomic units of scientific knowledge. A proposition differs from a

sentence: the same proposition may be expressed in different languages, different formalisms, or at different levels of abstraction.

Definition 2 (Provenance Triple). For any proposition p in a knowledge base K , the provenance triple is:

$$\text{prov}(p) = (\text{attribution}, \text{evidence}, \text{derivation})$$

where: - *attribution* identifies the source: who stated p , in what work, when. - *evidence* characterizes the support: type (formal proof, empirical, conceptual), strength, and references to the evidence base. - *derivation* specifies how p follows: from which prior propositions, by which logical or mathematical method.

A proposition without a complete provenance triple is an unverified claim — indistinguishable in principle from noise. The triple is the minimum structure that makes a claim epistemically tractable.

Definition 3 (Confidence Score). The confidence score $C_1(p) \in [0, 1]$ of a proposition p is a measure of the evidential support for p in the collective knowledge base. It is computed from the provenance triple, the number of independent derivations, and the confidence of the base propositions from which p is derived.

Formally, if p has n independent derivations from bases $B_1, \dots, B_n$:

$$C_1(p) = 1 - \prod_{i=1}^n (1 - C_1(B_i) \cdot s_i)$$

where $s_i \in [0, 1]$ is the proof strength of derivation i (1 for formal proof, decreasing for empirical, theoretical, conceptual). Independence of derivations is required: derivations from the same evidence chain do not compound.

This formula expresses: the probability that p is false is the product of the probabilities that each independent derivation fails. As independent derivations accumulate, $C_1(p)$ approaches 1. A single derivation with $s = 1$ and $C_1(B) = 1$ gives $C_1(p) = 1$ (certain). A single formal proof from a well-established base gives high but not certain confidence.

Note on the specific formula: We propose this confidence score as one computable approach to quantifying evidential support. Other formulations — Bayesian updating, Dempster-Shafer belief functions, calibrated prediction markets — may prove more appropriate for specific domains or evidence types. The contribution is the *principle* (that confidence should be computed from provenance and independent derivation, not inherited from citation count) rather than the specific formula, which we offer as a concrete starting point for the community to evaluate, critique, and improve.

Critical property: $C_1(p)$ is computed, not inherited. A claim cited 1,000 times by papers that cite each other is not 1,000 independent derivations — it is one derivation propagated 1,000 times. The current citation system conflates propagation count with independent derivation count. $C_1(p)$ does not.

2.2 Equivalency Classes

Definition 4 (Equivalency Class). An equivalency class EC is a set of propositions $\{p_1, p_2, \dots, p_k\}$ drawn from k distinct frameworks $\{F_1, \dots, F_k\}$ such that for all i, j : p_i and p_j

express the same underlying constraint on the domain, and a proof of p_i in F_i yields a proof of p_j in F_j under formal translation.

Equivalency classes are the cross-framework analog of a normal form: they identify the shared base assumptions on which multiple independent frameworks converge. An equivalency class with members from $k \geq 3$ independent traditions has high $C_1(p)$ because it requires k independent failures to invalidate.

Remark (Levels of equivalence). Equivalency classes admit degrees of strength: - *Formal equivalence*: $p_i \Leftrightarrow p_j$ under a formal translation, with proof. - *Structural equivalence*: p_i and p_j describe the same structural constraint at different levels of abstraction (physical, statistical, cognitive, philosophical). - *Conceptual equivalence*: p_i and p_j express the same intuition but resist formal translation. Conceptual equivalence should be marked explicitly and treated with lower confidence than formal equivalence.

A note on the interpretive challenge: Constructing equivalency classes requires judgment — deciding that two propositions from different frameworks express the same underlying constraint is an interpretive act. We have found the three-grade taxonomy (formal, structural, conceptual) useful as a starting point, but we do not claim it is complete. Different researchers applying this method to the same set of frameworks may produce different EC assignments, particularly at the structural-conceptual boundary. We view this as a feature, not a bug: the disagreements between independent EC assignments are themselves informative, and resolving them is exactly the kind of cross-framework dialogue this paper seeks to encourage. The goal is community consensus on what counts as equivalent, built through iterative refinement rather than imposed by any single analysis.

2.3 The Decomposition Theorem

Theorem (Decomposition). Let $\mathcal{F} = \{F_1, \dots, F_n\}$ be a set of frameworks sharing a base EC . Let q be a proposition derivable from EC in framework F_i . Then exactly one of the following holds:

- (a) **Base**: $q \in EC$ — q is a shared base assumption, not a contribution of F_i .
- (b) **Collapse**: $q \notin EC$, and there exists $F_j \neq F_i$ such that q is derivable from EC in F_j without contradiction. F_i and F_j independently confirm q . Each confirmation raises $C_1(q)$.
- (c) **Contradiction**: $q \notin EC$, and there exists $F_j \neq F_i$ such that $\neg q$ (or a proposition inconsistent with q) is derivable from EC in F_j . Either F_i or F_j is wrong about q , or their scopes differ in a way that must be made explicit.
- (d) **Novel**: $q \notin EC$, and q is not derivable from EC by any $F_j \neq F_i$, and q is consistent with all claims derivable from EC in any F_j . F_i has produced a genuine contribution not present in any prior framework.

Proof sketch. The four cases are mutually exclusive by construction. (a) covers $q \in EC$. For $q \notin EC$: either there exists F_j that derives q (collapse) or derives $\neg q$ (contradiction) or neither (novel). Exactly one holds. $\square$

Corollary (Peer review insufficiency). Standard peer review checks: (i) novelty relative to cited literature, and (ii) internal consistency. It does not check: whether q is a collapse relative to all F_j that share EC (because the reviewer checks only the cited literature, not all frameworks that share EC), whether q contradicts a claim in some uncited F_j that shares EC , or what the $C_1(q)$

of the novel claims is. Peer review is therefore necessary but not sufficient for the epistemic claims it is asked to validate.

2.4 The Intersection Prediction

Proposition (Emergent knowledge). Let K_i and K_j be the proposition sets of two frameworks sharing base EC . The intersection $K_i \cap K_j$ (confirmed overlaps) and the symmetric difference $K_i \triangle K_j$ (distinct contributions) together contain propositions EM not derivable from K_i alone or K_j alone, producible only by holding both frameworks simultaneously:

$$EM = \{q : q \text{ derivable from } K_i \cup K_j, q \notin K_i, q \notin K_j\}$$

These emergent propositions are new knowledge produced by the intersection. They are not present in any individual framework. They arise from the juxtaposition.

This is the prediction: the intersection of sufficiently distinct frameworks should produce emergent propositions. Section 5 demonstrates three such propositions arising from the running example.

3. Knowledge Graph Schema

The framework is implemented as a directed graph with typed nodes and provenance-carrying edges.

3.1 Node Types

Type	Meaning	C_1(p) basis
definition	Formal definition within a framework	Framework-internal
theorem	Proved theorem	Formal proof
reference	External framework proposition	Attribution + citation
overlap	Proposition grounded in 2+ frameworks independently	Multiple independent derivations
novel	Proposition grounded only in one framework	Single derivation
emergent	Proposition produced by intersection; not in any individual framework	Intersection of 2+ projections
equivalency	Equivalency class: shared base assumption across frameworks	k independent groundings

3.2 Edge Types

Every edge carries a provenance triple (attribution, evidence, derivation):

Relation	Meaning
<code>derives_from</code>	This proposition follows from the target
<code>overlaps_with</code>	Same underlying claim in both nodes, independently grounded
<code>generalizes</code>	This node generalizes the target (target is a special case)
<code>grounds</code>	This node provides formal grounding for the target
<code>qualifies</code>	This node adds a scope restriction to the target
<code>contradicts</code>	This node conflicts with the target from shared premises
<code>emergent_from</code>	This node was produced by the intersection of target nodes
<code>philosophical_precursor</code>	Target is a qualitative predecessor of this formal result

3.3 The Provenance Invariant

Invariant: Every node and every edge in the graph must have a complete provenance triple. A node or edge without provenance is structurally indistinguishable from noise.

This invariant operationalizes the confidence chain: every claim in the graph is traceable to its evidence base. Removing a node ripples through to every node that cites it. The revalidation gap (Section 1) becomes visible: if [A]’s evidence for P weakens, the graph shows every edge that depends on [A]’s grounding of P.

4. Demonstration: Evaluating a Formal Theory of Intelligence

We apply the framework to a formal theory of intelligence — *Uncertainty Bounding as the Basis of Intelligence* [McCarthy2025] — evaluated against fifteen prior frameworks spanning formal AI theory, cognitive neuroscience, information theory, philosophy of science, and thermodynamics.

4.1 The Target Theory

The uncertainty bounding (UB) framework proposes that intelligence is the driven capacity to improve an entity’s action space relative to a world it cannot fully represent [McCarthy2025]. Its core claims:

- $I(E) = \mathcal{A} \cdot \eta_M$: Intelligence is the product of agency $\mathcal{A} \in [0, 1]$ (the probability that the entity engages its learning mechanism in response to a gradient signal) and η_M (the rate at which the learning mechanism improves the action space per unit of learning signal).
- $M < N$ **always**: The entity senses M dimensions of the world W at any moment; W has N dimensions. $M < N$ is formally derived from bounded context C_n and execution dynamics.
- **Graph necessity**: At the bounded context limit C_n , factoring is the unique information-preserving compression strategy. Graph construction is necessary, not chosen.

- **Intersection acceleration:** The intersection of independently-derived knowledge graphs accelerates the rate of $M \rightarrow N$ convergence by providing confirmed overlap propositions (higher $C_1(p)$) and distinct dimension extension (more M per operation).

4.2 The Reference Frameworks

Fifteen frameworks were represented as graph nodes with their key propositions enumerated:

Formal AI and cognitive theory: Schmidhuber [2010] (compression-progress formal theory), Friston [2010, 2017, 2021] (free energy principle; active inference; sophisticated inference), Thrun and Pratt [1998] (meta-learning), Tishby and Polani [2011] (information bottleneck).

Information theory: Shannon [1948] (entropy and communication), Cover and Thomas [2006] (elements of information theory including the data processing inequality and chain rule).

Cognitive science: Baars [1988] (global workspace theory), Dehaene and Changeux [2011] (neural global workspace theory), Kahneman [2011] (dual-process theory).

Philosophy: Peirce [1877, 1878] (pragmatist epistemology: fixation of belief, pragmatic maxim), Tarski [1936] (semantic theory of truth and undefinability), Chalmers [1995] (hard problem of consciousness — for scoping).

Thermodynamics and biology: Schrödinger [1944] (negentropy and life), Prigogine and Stengers [1984] (dissipative structures), England [2013] (statistical physics of self-replication), Maynard Smith and Szathmáry [1995] (major transitions in evolution).

For each framework, key propositions were enumerated and entered as reference nodes with provenance triples. Cross-framework edges were added where a proposition in one framework grounds, overlaps with, or contradicts a proposition in another.

5. Results

5.1 Equivalency Classes

Analysis of the full reference graph identified ten equivalency classes — shared base assumptions on which multiple independent frameworks converge:

EC01 (Bounded context, $C_1 = 0.92$). C_n (UB) $\equiv$ approximate posterior (Friston) $\equiv$ global workspace capacity (Baars, Dehaene) $\equiv$ finite deliberate attention (Kahneman). Four traditions from formal AI, neuroscience, and psychology independently converge on a bounded internal representation capacity.

EC02 (Rate of model improvement, $C_1 = 0.88$). η_M (UB) $\equiv \frac{d}{dt}[\text{compression}(C)]$ (Schmidhuber) $\equiv$ rate of free energy reduction (Friston) $\equiv$ meta-learning rate (Thrun). The derivative of model quality — not absolute quality — is independently identified as the measure of intelligence by four distinct traditions.

EC03 (Structured representation, $C_1 = 0.85$). K/F graph (UB) $\equiv$ hierarchical generative model (Friston) $\equiv$ compression (Schmidhuber, implied) $\equiv$ sufficient statistic (Cover, Tishby) $\equiv$ affordance structure (Gibson). Five traditions agree that the representation must be structured. UB additionally proves the graph form is *necessary*; the others ground that structure is *required*.

EC04 (Survival threshold, $C_1 = 0.94$). U_{lethal} (UB) $\equiv$ Markov blanket maintenance threshold (Friston) $\equiv$ thermodynamic self-replication threshold (England) $\equiv$ negentropy floor (Schrödinger) $\equiv$ dissipative structure bifurcation point (Prigogine). Five traditions from cognitive science, statistical mechanics, and thermodynamics derive the same physical threshold.

EC05 (Epistemic incompleteness, $C_1 = 0.97$). $M < N$ (UB) $\equiv q(s) \neq p(s)$ (Friston, Jensen’s inequality) $\equiv$ truth as long-run convergence (Peirce) $\equiv$ undefinability (Tarski) $\equiv$ incompleteness (Gödel, via Tarski). This is the highest-confidence equivalency class: formal mathematical proofs from multiple independent traditions (incompleteness theorems, Jensen’s inequality) establish that no physical system can fully represent the world from which it emerged.

EC06 (Belief as action disposition, $C_1 = 0.95$). K grounds F (UB) $\equiv$ belief as habit of action (Peirce, pragmatic maxim) $\equiv$ posterior drives action (Friston, active inference) $\equiv$ affordance perception (Gibson) $\equiv$ global broadcast enabling action (Baars). Five traditions independently establish that retained knowledge has value only through its effect on action. The UB grounding relation formally instantiates Peirce’s 1878 pragmatic maxim.

EC07 (Collective knowledge integration, $C_1 = 0.83$). $K_{collective}$ (UB) $\equiv$ major transitions in evolution (Maynard Smith) $\equiv$ community of inquiry (Peirce). Three traditions ground that collective knowledge integration produces qualitatively superior representations. The evolutionary and philosophical traditions provide empirical and conceptual grounding; UB provides the formal mechanism (Section 5.3).

EC08 (Compression through factoring, $C_1 = 0.91$). Factoring at C_n (UB) $\equiv$ compression progress (Schmidhuber) $\equiv$ chain rule decomposition (Cover) $\equiv$ information bottleneck (Tishby) $\equiv$ aperiodic encoding (Schrödinger). Five traditions establish that retaining more in less space requires extracting shared structure. UB’s unique contribution: proving the graph form is *necessary*, not just that structure is needed.

EC09 (Hierarchical cost triage, $C_1 = 0.90$). L0–L3 hierarchy (UB) $\equiv$ System 1/System 2 (Kahneman) $\equiv$ global workspace levels (Baars, Dehaene) $\equiv$ precision hierarchy (Friston). Four traditions converge on the same hierarchical routing structure: cheaper more-certain processing is handled locally; more expensive uncertain processing escalates.

EC10 (Uncertainty as inquiry driver, $C_1 = 0.93$). $U(w, K) > 0$ (UB) $\equiv$ variational free energy > 0 (Friston) $\equiv$ doubt (Peirce) $\equiv$ compression gap (Schmidhuber) $\equiv$ cognitive dissonance (Kahneman). Five traditions independently identify the gap between current representation and world structure as the driver of intelligent behavior. Given EC05 ($M < N$ always, $C_1 = 0.97$), this gap is always positive: the driver never terminates.

5.2 Confirmed Overlaps (OV Nodes)

Four propositions were identified as confirmed precipitation sites — grounded by two or more frameworks independently:

OV01: Bounded internal world representation (UB + FEP + GWT). **OV02:** Rate of model improvement as intelligence measure (UB + Schmidhuber). **OV03:** Non-termination of learning (UB formal; FEP informal under non-stationarity). **OV04:** Meta-cognitive awareness of own inference process (UB + Friston 2021).

These propositions have $C_1(p)$ elevated above any single framework by their independent grounding.

5.3 Novel Contributions (NV Nodes)

Five propositions were identified as grounded only in UB — not derivable from the shared equivalency class base by any other framework in the reference set:

NV01 (Graph structure is necessary, not sufficient). EC08 establishes that structured compression is needed. UB’s Theorem 2b Part 3 adds: the graph form specifically is the *unique* information-preserving option at C_n , proven by the two-option argument (discard without structure $\Rightarrow$ information loss by the data processing inequality [Cover2006]; factor $\Rightarrow$ graph construction $\Rightarrow$ information preservation). No prior framework derives this necessity. *This is UB’s strongest individual contribution.*

NV02 ($M < N$ formal mechanical derivation). EC05 establishes incompleteness with $C_1 = 0.97$. UB’s Theorem 3 Part II adds: the mechanism by which $M < N$ is maintained — C_n bounds context at each step; execution extends $\mathcal{T}_{reachable}$; for $\mathcal{A} > 0$, there always exist reachable states not yet covered by K . Gödel and Tarski prove incompleteness by logical necessity; UB proves it by physical mechanism.

NV03 (Agency as engagement probability; the $\mathcal{A} = 0$ case). All prior frameworks either implicitly assume $\mathcal{A} = 1$ (Schmidhuber: the agent always maximizes compression progress) or define the agent by its engagement (Friston: the Markov blanket is the agent boundary). UB parameterizes $\mathcal{A} \in [0, 1]$ and derives that $\mathcal{A} = 0$ entities are removed by selection pressure (Theorem 6 Part 1). This is the formal account of why prior frameworks can assume engagement: entities with $\mathcal{A} = 0$ do not persist to be studied.

NV04 (Intersection formally accelerates $M \rightarrow N$). EC07 establishes that collective knowledge integration produces qualitative transitions. UB’s Corollary 8a provides the mechanism with cardinality: solo execution adds one dimension per step; intersection with K_j adds $|K_j \setminus K_i|$ dimensions per operation. The formal cardinality argument is unique to UB.

NV05 (Higher-order sentience is derived, not axiomatic). OV04 establishes that meta-cognitive awareness is independently grounded. UB derives *why* it is adaptive: from factorization $\rightarrow$ compactness $\rightarrow |K|$ growth (Theorem 2b Part 3) + $M < N$ (Theorem 3 Part II) + retention criterion (Theorem 1), improving the normalization process is strictly dominant across all future survival pressures. Prior frameworks make meta-cognition architectural; UB makes it derivable.

5.4 Emergent Propositions (EM Nodes)

Three propositions were identified that do not appear in any individual framework but arise from holding multiple frameworks simultaneously:

EM01: Schmidhuber’s formal agent is UB with $\mathcal{A} = 1$. NV03 ($\mathcal{A}$ as probability gate) applied to Schmidhuber’s implicit $\mathcal{A} = 1$ (S4) produces a containment claim: Schmidhuber’s framework is a special case of UB where engagement is certain. Neither Schmidhuber (who has no $\mathcal{A}$ parameter) nor UB (which does not explicitly identify Schmidhuber as a special case in its text) states this. It is produced by the intersection. Consequence: UB generalizes Schmidhuber in two formal directions — partial engagement and graph form necessity.

EM02: FEP’s equilibrium claim requires the $M < N$ stationarity qualification. FEP6 (free energy approaches its minimum) and Theorem 3 Part II ($M < N$ always) share base assumption EC05 (incompleteness). From EC05, $M < N$ is a structural consequence, not an empirical

observation. FEP6’s equilibrium requires a fixed, stationary environment — a condition not embedded in FEP’s assumptions. UB’s $M < N$ derivation makes this condition explicit: FEP6 is the fixed-world idealization; in open environments, $M < N$ is maintained. Neither FEP nor UB states this scope qualification explicitly; it emerges from their intersection.

EM03: UB’s necessity proof provides the derivation FEP’s architectural assumption requires. FEP7 (hierarchical graph-structured generative models are assumed) and NV01 (graph structure is proved necessary at C_n) share base assumption EC01 (bounded context) and EC08 (compression through factoring). The intersection reveals mutual support: NV01 provides the derivation FEP7 needs (why the architecture must be hierarchically structured); FEP’s empirical and theoretical success provides evidence that graph-structured representations work (supporting NV01’s formal necessity claim). Neither framework states the bilateral grounding relationship; it is produced by the intersection.

Confirmation of the intersection prediction (Section 2.4): three emergent propositions were produced, confirming that the intersection of sufficiently distinct independent projections generates new knowledge not present in any individual framework.

6. Theory Confidence Profile

Applying the Decomposition Theorem (Section 2.3) to UB’s major results:

Result	Classification	$C_1(p)$	Primary basis
T1 (Minimization Imperative)	COLLAPSE	0.93	EC04 + EC06 + EC10
T2b Part 3 (Graph necessity)	NOVEL	0.82	EC08 base + novel two-option proof
T3 Part I (Convergence)	COLLAPSE + NOVEL (structure)	0.87	EC10 + optimization theory
T3 Part II ($M < N$)	COLLAPSE + NOVEL (mechanism)	0.95	EC05 (Gödel/Tarski)
T6 ($\mathcal{A} > 0$ necessity)	NOVEL	0.80	No prior framework
T6 ($\eta_M > 0$ necessity)	COLLAPSE	0.90	EC02 + EC04
C6a ($I = \mathcal{A} \cdot \eta_M$)	NOVEL (requires NV03)	0.90	Law of total expectation
C8a (Intersection acceleration)	NOVEL	0.78	EC07 qualitative; cardinality novel
D14 (Sentience)	COLLAPSE + NOVEL (derivation)	0.85	OV04 + NV05

No contradictions were found. Two scope mismatches were identified and resolved: FEP6’s equilibrium (resolved as stationarity qualification, EM02) and Gibson’s direct perception (resolved by

scope: perceptual coupling vs. knowledge accumulation over time). The theory is internally consistent with all prior frameworks in the reference set.

The theory’s weakest claims ($C_1 < 0.82$) are precisely its most novel ones: graph form necessity (T2b Part 3) and cardinality acceleration (C8a). This is structurally expected: novel claims have fewer independent derivations. Both are empirically testable:

- *Graph form necessity*: compare information retention of graph-structured vs. non-graph agents under bounded context constraints. Prediction: non-graph agents lose information at C_n ; graph agents do not. This distinguishes UB from Schmidhuber’s unconstrained C .
- *Cardinality acceleration*: measure effective dimension coverage per operation in multi-agent vs. solo learning systems. Prediction: intersection provides $|K_j \setminus K_i|$ dimensions per operation; solo execution provides 1.

7. A Proposal for Explicit Epistemic Provenance in Peer Review

The analysis establishes what the current review process does not check. We want to propose one possible standard that could extend peer review to address these gaps — offered in the spirit of starting a conversation about what the right extensions would look like, not as a prescriptive mandate. The specific requirements below are our best attempt at concreteness; we expect and hope they will be refined, adapted, and in some cases replaced by better alternatives as the community engages with them.

7.1 The Current Standard and Its Gaps

Peer review currently validates: 1. *Novelty*: is the contribution new relative to the cited literature? 2. *Internal consistency*: are the proofs correct, the experiments valid, the claims supported by the paper’s own evidence? 3. *Significance*: is the contribution important enough for the venue?

It does not validate: 4. *Collapse detection*: is the claimed contribution actually derivable from equivalency classes shared with uncited frameworks? (A result that collapses into prior work from a shared base is a confirmation, not a novel contribution — even if the paper is unaware of the prior framework.) 5. *Contradiction checking*: does the paper derive claims that contradict results in frameworks it did not cite, from shared base assumptions? 6. *Confidence chain tracing*: what is the $C_1(p)$ of each cited claim in its original source, and is the current paper’s use of that claim consistent with its evidential scope? 7. *Emergent proposition identification*: does the paper, in combination with prior work, produce propositions that should be identified and credited?

Gaps 4–7 are not failures of reviewers; they are failures of the current structure. No individual reviewer can check a paper against all frameworks sharing its equivalency classes — because those equivalency classes have not been enumerated and the relevant frameworks have not been identified. The knowledge graph makes this tractable.

7.2 The Proposed Standard

We propose the following additions to the peer review process:

R1 (Equivalency class declaration). Authors declare which equivalency classes their base assumptions belong to. Reviewers check whether the declared equivalency class members are correct

and complete. This is analogous to related work, but scoped to base assumptions rather than prior results.

R2 (Collapse/novel/contradiction classification). For each major claim, authors classify it as: (a) a collapse — an independently-confirmed result from shared premises, with citation of the independent derivation; (b) a novel contribution — not derivable from the EC base by any cited framework, with explicit statement of the gap; or (c) a resolved contradiction — an apparent conflict with a prior framework, with the resolution stated.

R3 (Confidence scores for novel claims). Novel claims (classification (b) above) include an explicit $C_1(p)$ estimate and the evidence basis. This makes the epistemic status of novel contributions transparent: a novel claim with $C_1 = 0.78$ (like UB’s C8a) is clearly distinguished from one with $C_1 = 0.95$ (like UB’s T3 Part II).

R4 (Provenance triples for cited claims). For each claim cited from prior work, authors provide the specific claim (not just the paper), the evidence type in the original source (formal proof, empirical, conceptual), and whether they have independently verified the claim or are relying on attribution.

R5 (Knowledge graph submission). The knowledge graph representation of the paper’s propositions and their provenance is submitted alongside the paper. This is machine-readable, enables automated collapse/contradiction checking, and accumulates into the collective knowledge graph of the field.

7.3 Feasibility and Adoption Path

The proposed standard adds work. We address the feasibility concern directly.

For authors: R1 and R2 require thinking that strong theoretical work already does implicitly. Making it explicit is additional writing, not additional research. R3 requires calibrating a confidence estimate — a discipline that formal work already exercises. R4 and R5 are partially automatable: reference managers can be extended to capture provenance triples; large language models can assist in extracting propositions from existing papers.

For reviewers: R1 and R2 reduce the review task — the reviewer is no longer asked to intuit novelty from a general reading but to check specific classifications against a declared equivalency class base. This is more tractable, not less.

For venues: adoption can be incremental. A venue can require R1 and R2 initially, adding R3–R5 as tooling matures. Venues focused on formal theory have the most immediate benefit; empirical venues follow as provenance tooling extends to experimental claims.

For the field: a submitted knowledge graph accumulates into the collective knowledge graph of the field. Over time, the equivalency class database grows, making automated collapse/contradiction checking faster. The marginal cost of R5 decreases as the graph grows.

8. Scaling and the Emergent Knowledge Prediction

8.1 From One Paper to the Full Literature

The running example evaluated one theory against fifteen frameworks. The framework scales to the full literature by the following structure:

Each paper is a projection K_i with M_i propositions. Applying the Decomposition Theorem across all pairs (K_i, K_j) produces: - A confidence map: each proposition's $C_1(p)$ computed from all independent derivations. - A collapse register: claims that appear novel in their original papers but are independently derived in other frameworks. - A contradiction register: claims that conflict with other frameworks from shared bases. - An emergent proposition set: new knowledge produced by the intersection.

The method's prediction: the emergent set grows with each new framework added to the graph. Each new framework K_n adds $|K_n \setminus K_{collective}|$ novel propositions and $|K_n \cap K_{collective}|$ confirmations. It also produces $|EM_n|$ emergent propositions from its intersection with $K_{collective}$ — propositions not in K_n or $K_{collective}$ individually, but visible once both are held simultaneously.

This is the prediction of the formal theory applied to itself: $K_{collective}$ is the maximal current measure of what the field knows. Each new paper extends it.

8.2 The Confidence Chain Audit

Scaling the provenance framework to the full literature produces what current review practice cannot: a confidence chain audit.

For any claimed fact in the literature, the audit traces: - Which original evidence base does this claim derive from? - How many independent derivations support it? - What is the $C_1(p)$ at each step in the citation chain? - Where does the chain terminate in primary evidence (experiment, formal proof) vs. propagated citation?

The audit makes visible what is currently invisible: foundational assumptions cited universally but grounded in a single original study; results with high citation counts but low $C_1(p)$ because all citations trace to one derivation; contradictions between frameworks that have never been cited together.

8.3 The Revalidation Signal

A key property of the knowledge graph: revalidation propagates automatically.

If a foundation node B is revised — a failed replication, a boundary condition discovered, an assumption tightened — the graph shows every edge that depends on B . The affected nodes are not automatically invalidated, but they are flagged for review. The current citation system has no such propagation: a published result citing a weakened foundation continues to circulate at its original implied confidence level indefinitely.

9. Discussion

9.1 Relation to Existing Scientometrics

Bibliometric analysis has studied citation patterns extensively [Garfield1955, Hirsch2005]. Citation networks have been used to identify influential papers, track idea propagation, and detect research communities. This work is valuable but operates at the level of attribution — it counts citations without examining what specific claims are being cited, how strong their evidence basis is, or whether they are consistent with the claiming framework’s other commitments.

Knowledge representation research (OWL, RDF, linked data) has developed formal schemas for expressing structured knowledge with typed relations [Bizer2009, Heath2011]. Biomedical knowledge graphs [Nicholson2020] have applied this to scientific claims. The contribution here is the provenance-first design: not just structured knowledge but knowledge with explicit confidence chains and the Decomposition Theorem as a tool for cross-framework analysis.

The philosophy of science has addressed related questions: Kuhn’s paradigm incommensurability [Kuhn1962], Lakatos’s research programs [Lakatos1978], and Popper’s falsifiability criterion [Popper1959] all address how scientific frameworks relate to each other and to evidence. The equivalency class formalism makes the question of framework comparison formally tractable where Kuhn and Lakatos left it qualitative.

9.2 Limitations

The framework requires upfront work to enumerate propositions and build the graph. For mature, large-scale fields, the initial construction is a significant undertaking. Automated proposition extraction from papers is an active research area [Kardas2020] but not yet reliable enough for unsupervised deployment.

The equivalency class construction requires judgment: identifying whether two propositions from different frameworks express the same underlying constraint is itself an interpretive act. We have distinguished three grades of equivalence (formal, structural, conceptual) and flagged conceptual equivalence as carrying lower confidence — but the boundary between structural and conceptual is not always sharp.

$C_1(p)$ is a model of confidence, not confidence itself. The formula in Section 2.1 makes specific assumptions about independence and proof strength that may not hold in all cases. It should be treated as a structured estimate, not a precise probability.

9.3 The Hard Question: What Does This Mean for Published Claims?

If the framework is applied retrospectively to the existing literature and finds that many widely-cited claims are single-derivation with $C_1(p) < 0.7$, while others with high citation counts are collapses of results proved elsewhere — what follows?

The answer is not that these claims are wrong. A single well-executed formal proof is sufficient to establish a proposition. High novelty + low $C_1(p)$ means the claim is specific, not that it is false. The framework’s output is not a verdict; it is a map. The map shows where the field’s knowledge is load-bearing vs. where it is thin; where independent confirmation would make the largest difference; where the next experiment or proof should go.

This is exactly what science is supposed to do. The current citation system makes this map invisible. The proposed framework makes it explicit.

10. Conclusion

The confidence chain problem is structural: scientific citation practice conflates attribution with confidence, implies revalidation that does not occur, and lacks a mechanism for cross-framework consistency checking. These are not failures of individual scientists or reviewers; they are absences in the current infrastructure.

We have proposed a formal solution: explicit provenance triples, computable confidence scores, equivalency classes as shared base assumptions, and the Decomposition Theorem as the tool for classifying every derivation as collapse, contradiction, or novel.

The demonstration on a formal theory of intelligence confirms: - Ten equivalency classes identified, ranging from $C_1 = 0.83$ to $C_1 = 0.97$. - Four confirmed overlap propositions with cross-framework support. - Five genuinely novel contributions, with $C_1(p)$ scores that distinguish the better-supported from the less-supported. - Three emergent propositions produced by cross-framework intersection — new knowledge not present in any individual framework.

The intersection prediction (Section 2.4) is confirmed: the intersection of sufficiently distinct frameworks produces new knowledge. The Decomposition Theorem classifies every proof without remainder.

The proposed peer review standard (Section 7.2) adds five requirements: equivalency class declaration, collapse/novel/contradiction classification, confidence scores for novel claims, provenance triples for cited claims, and knowledge graph submission. The standard is incrementally adoptable and automatable.

The deepest implication is the simplest: the confidence that scientific results carry is currently implied, inherited via citation, and never explicitly stated. Making it explicit does not change what scientists do — it makes visible what they have always been doing, and reveals where the foundations are strong and where they are thin.

We offer this framework in the belief that the scientific community, which has built the most extraordinary knowledge-producing system in human history, can make that system even better by making its implicit practices explicit. The specific proposals in this paper are a starting point. The conversation about what the right practices should be is the contribution we most hope to make. We look forward to being corrected, improved upon, and composed with — because that is, after all, what the framework predicts should happen.

Acknowledgments

The formal theory evaluated in Section 4 is the author’s own prior work. We chose to evaluate our own framework as the running example because it is the framework we have the most complete access to — every proposition, every derivation, every assumption is known to us at full depth. This is both a strength (the demonstration is thorough) and a limitation (the evaluation is not independent). We view the most valuable next step as independent researchers applying the

method to frameworks they know well, using the same protocol, and comparing their results to ours. The method is designed to be applied by anyone to any framework, and its value will be best demonstrated by multiple independent applications rather than by the single evaluation presented here. We warmly invite this.

The knowledge graph implementation and cross-framework analysis described in this paper are available at <https://github.com/patdmc/open-knowledge-graph>.

References

The following references are cited above. Full bibliographic details are in `references.bib` in the accompanying repository.

- Baars1988: A Cognitive Theory of Consciousness
- Chalmers1995: Facing Up to the Problem of Consciousness
- Cover2006: Elements of Information Theory
- Dehaene2011: Experimental and Theoretical Approaches to Conscious Processing
- England2013: Statistical Physics of Self-Replication
- Friston2010: The Free-Energy Principle
- Friston2017: Active Inference: A Process Theory
- Friston2021: Sophisticated Inference
- Gibson1979: The Ecological Approach to Visual Perception
- Kahneman2011: Thinking, Fast and Slow
- MaynardSmith1995: The Major Transitions in Evolution
- McCarthy2025: Uncertainty Bounding as the Basis of Intelligence [companion paper]
- Peirce1877: The Fixation of Belief
- Peirce1878: How to Make Our Ideas Clear
- Prigogine1984: Order Out of Chaos
- Schmidhuber2010: Formal Theory of Creativity, Fun, and Intrinsic Motivation
- Schrodinger1944: What is Life?
- Shannon1948: A Mathematical Theory of Communication
- Tarski1936: The Concept of Truth in Formalized Languages
- Thrun1998: Learning to Learn
- Tishby2011: Information Theory of Decisions and Actions